Historical Inverse PINN Study of Lynx and Hare Pelt Dynamics

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1 Historical inverse PINN study

This is Yash's contribution, using the historical pelt series supplied by Advay. It extends the separately preserved synthetic recovery experiment; its individual scope is retained alongside the combined group deliverables. The Stan compilation contains annual Hudson's Bay Company hare and lynx pelt returns [1]. We train on 1900–1915 (16 pairs) and evaluate once on frozen forecasts for 1916–1920 (five pairs). Counts are thousands of pelts, treated as abundance indices rather than censuses. Annual totals are approximated as point samples. No observation scale factor is fitted. Alternative source compilations disagree; all results here are conditional on the Stan series.

1.1 Model and protocol

With $t = \text{Year} - 1900$, the shared dimensional model is

$$\dot{H} = rH(1 - H/K) - aHP/(b + H), \qquad \dot{P} = eaHP/(b + H) - mP.$$

All six positive coefficients are inferred. Bounds below are exploratory numerical choices, not measured biological ranges. No coexistence constraint is imposed and e is not constrained below one because these are effective coefficients in pelt units. The starting vector is $(r, K, a, b, e, m) = (0.8, 150, 2, 30, 0.6, 0.6)$. The shared historical configuration records this protocol and its exploratory bounds. The local least-squares benchmark uses the identical config, data split, scaling and solver.

Coefficient	Bounds	Selected primary estimate
r	0.05–3	0.60546
K	20–500	497.6
a	0.05–10	9.2605
b	0.5–300	299.98
e	0.02–5	1.0966
m	0.05–3	0.99388

1.2 Network, loss and optimization

The inverse PINN [2] uses three hidden layers of 48 tanh units, float64, softplus-positive normalized outputs, and sigmoid-bounded coefficients. Species scales are training maxima $(s_H, s_P) = (77.4, 59.4)$; network time is $2t/15 - 1$. For $u_\theta = (H_\theta/s_H, P_\theta/s_P)$, automatic differentiation evaluates $R_\theta = \partial_t u_\theta - f(u_\theta \odot s; p)/s$. The loss is

$$L = \text{mean}(u_\theta - y/s)^2 + 10\,\text{mean}\,R_\theta^2 + 10\,\text{mean}(u_\theta(0) - y_0/s)^2.$$

The initial observation is $y_0 = (30, 4)$. There are 161 collocation times in $[0, 15]$. Optimization uses 500 data/IC warmup steps, up to 5000 joint Adam steps at learning rate 0.002 (gradient

norm clipped at 10), then up to 600 L-BFGS iterations with strong-Wolfe line search. Training loss is logged every 100 Adam steps; plateau stopping uses 15 checks and minimum improvement 10^{-8}. The best recorded weighted training objective determines the saved state. Seeds are 17, 29 and 43, with one CPU thread and deterministic Torch operations. Each variant selects its own lowest-loss seed; objectives from different observation models are not used to select the primary analysis.

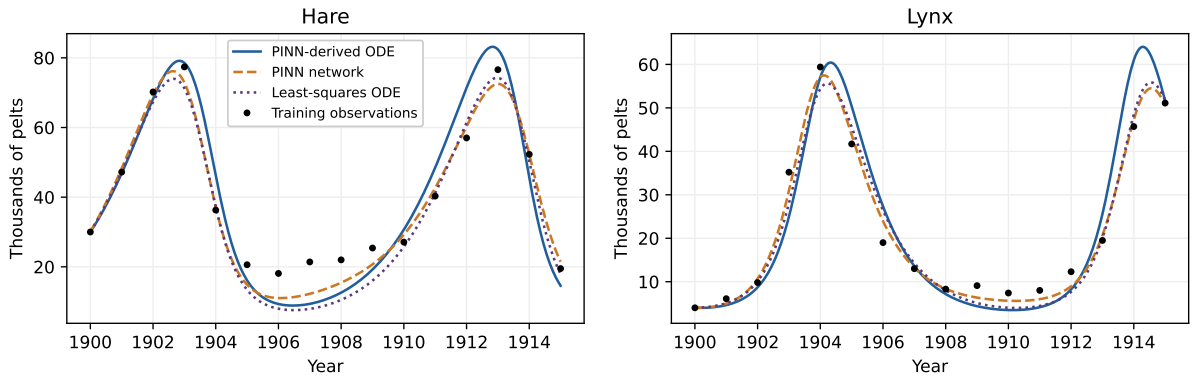


Figure 1: Training observations, learned network and re-integrated ODE. A small local physics residual does not guarantee identical long-horizon ODE and network trajectories.

1.3 Frozen forecasts and matched benchmark

The selected primary seed is 17. Forecasts use the learned coefficients in the shared RK45 solver with rtol 10^{-8}, atol 10^{-10} and maximum step 0.1 year. Integration starts from $(30, 4)$ in 1900 and proceeds through 1920, with no restart from a test state. The neural network is never extrapolated for forecasting. Its starting-state discrepancy is $(0.10060, -0.00723)$ thousand pelts; the ODE still starts at the exact observation. The primary ODE training normalized MSE is 0.00981. The matched primary least-squares fit uses six starts (RNG seed 91), bounded `least_squares`, up to 500 function evaluations and ftol=xtol=gtol=10^{-10}. Each start and its convergence status is retained. This independent benchmark was subsequently reproduced by Advay on the same configuration; the combined report documents the chronology. PINN and least squares differ in soft versus exact IC enforcement and in the PINN's added physics penalty.

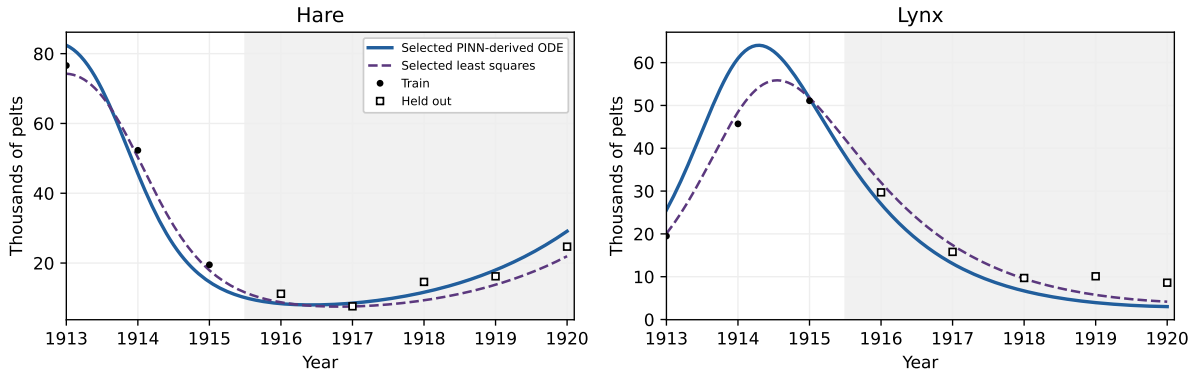


Figure 2: Frozen forecasts. Thin pale lines are the three primary PINN seeds, not confidence bands. Shading marks held-out years; observations were used for scoring only after all forecast and seed choices were frozen.

Method	RMSE		MAE	
	Hare	Lynx	Hare	Lynx
PINN fitted IC	3.687	4.697	3.057	4.530
PINN log errors	18.351	10.026	14.289	8.972
PINN primary	2.847	4.317	2.590	4.046
PINN wider bounds	2.976	4.053	2.721	3.779
PINN physics 100	4.519	3.978	4.107	3.496
LS primary	3.072	3.043	2.586	2.559
LS log errors	6.874	1.596	6.626	1.381
LS fitted IC	2.246	3.236	2.083	2.713

All scores are thousands of pelts on the same five held-out years. The primary PINN has slightly lower hare RMSE than primary least squares, but higher lynx RMSE. Hare MAE is essentially tied. These results do not establish an overall PINN advantage.

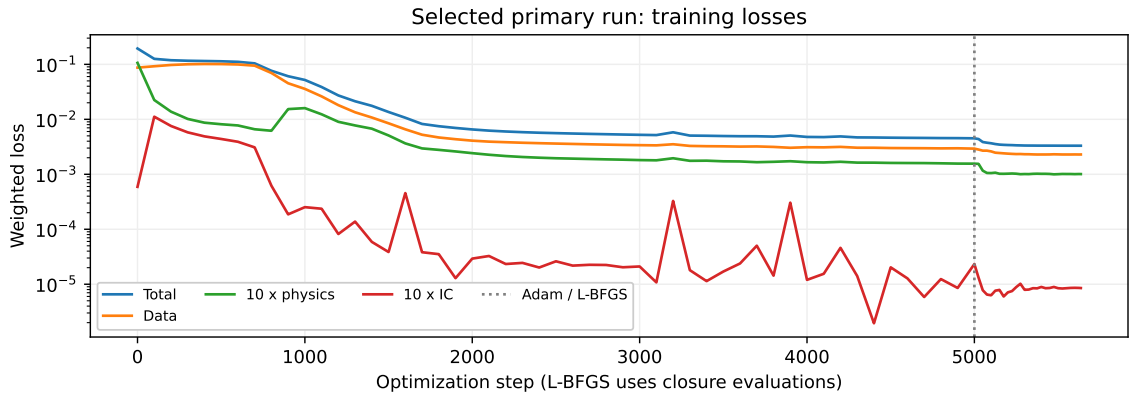


Figure 3: Selected primary training loss components. L-BFGS abscissae count closure evaluations after Adam; they are not directly comparable computational steps.

1.4 Sensitivity and practical identifiability

All sensitivities were specified before any test scoring and are reported regardless of outcome. The main three seeds were repeated with log observation residuals, and separately with trainable initial states in $[0.5, 1.5]$ times the 1900 observations. The first data point stays in the data loss. The fitted-IC PINN's ODE starts at approximately $(34.7477, 3.7841)$; the network and ODE initial states are separately recorded. The benchmark repeats both observation/IC sensitivities with six starts each. Log residuals worsen hare forecast error for both methods, especially the PINN. None of these scores changes the predeclared primary choice.

To check arbitrary box limits, each lower bound was halved and each upper bound doubled, and the main PINN was repeated for all three seeds. A further seed-17 run increased physics weight from 10 to 100. In total there are 13 PINN runs and 18 least-squares starts. The wider-box fits move K from about 498 to about 986 and b from about 300 to about 600, close to the expanded limits; a also shifts substantially. Similar results across seeds therefore do not demonstrate uniquely recovered biological parameters. The wider-box forecast changes are smaller than these coefficient shifts.

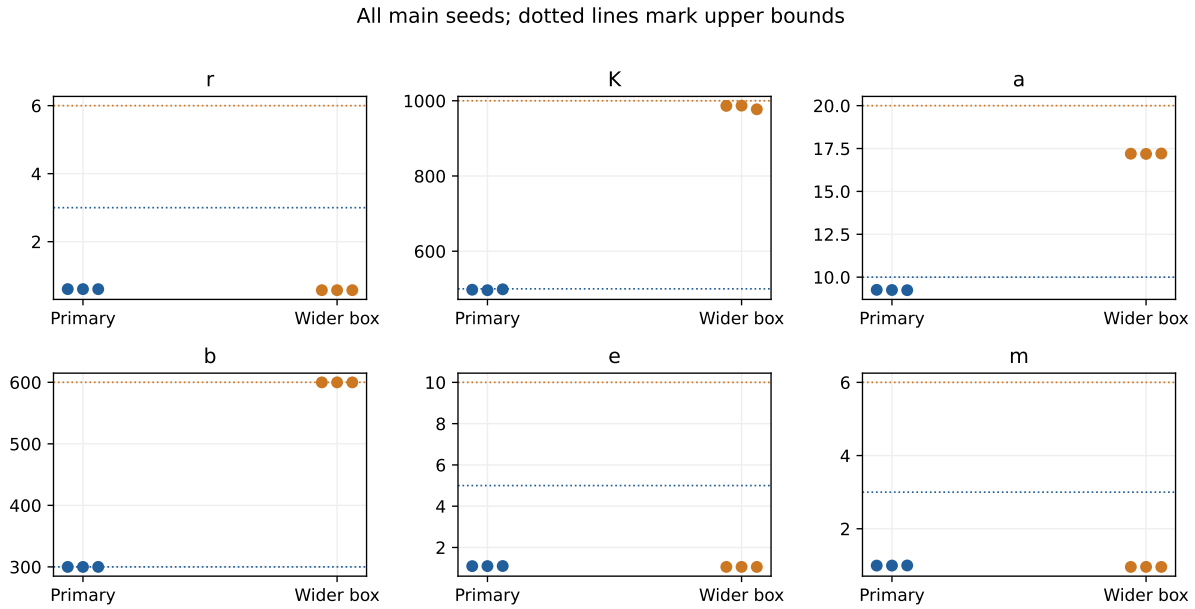


Figure 4: Primary versus wider-box parameter estimates over three seeds. Several effective coefficients track the numerical bounds; narrow seed spread is not parameter certainty.

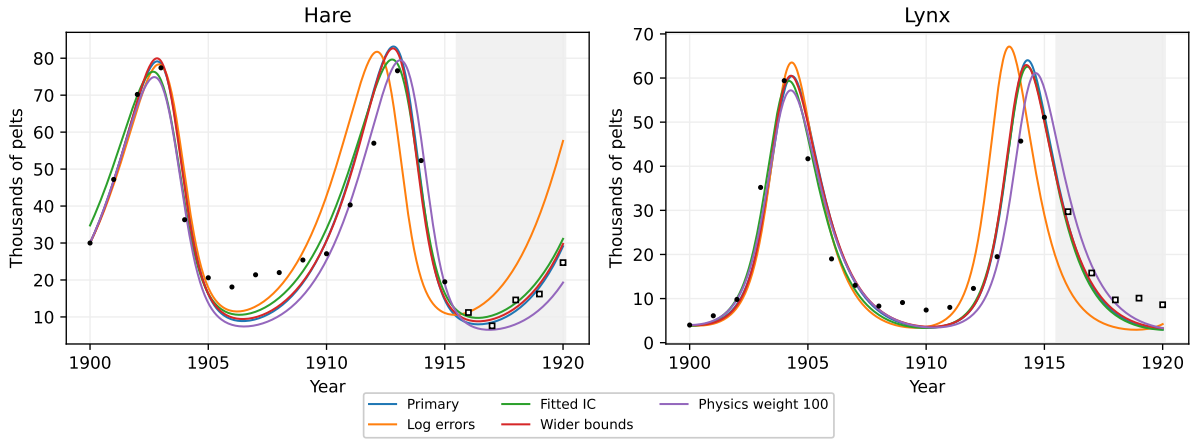


Figure 5: All selected PINN variants. The log-error fit shows appreciable model/ODE forecast sensitivity. Variants were retained rather than chosen using held-out errors.

The local trajectory sensitivity matrix uses central differences with respect to log parameters (step 10^{-4}), normalized using the declared observation loss. Its selected primary condition number is approximately 317.5. This is a local diagnostic, not a proof of structural identifiability or a confidence interval. For feasible coexistence, $H^* = mb/(ea - m)$, $K_H = b + 2H^*$ and $\sigma = K/K_H$; the selected primary fit gives $\sigma = 1.363$. This fitted ODE lies on the cycling side of Joel's threshold, conditional on the model and protocol. It does not prove that the historical ecological system has a stable limit cycle. All seeds' values and feasibility flags are exported to the team.

1.5 Reproducibility and limitations

`configs/historical.json` records every primary choice; the two sensitivity configs record the expanded box and physics weight. Run manifests hash sources, config, training data, forecasts and diagnostics. A forecast-freeze record is written before the test observations are read. `scripts/run_historical_pinn.py` accepts only the training table; `scripts/score_historical.py` handles final scoring. Existing synthetic recovery results are preserved as a separate controlled validation.

Six effective coefficients are fitted from roughly two noisy cycles; boundary hits, source disagreement, annual season totals, unknown catchability and the small test window limit interpretation. PINN physics is a soft constraint, and the log-loss variant exposes an appreciable network/ODE discrepancy. Seed spread is optimization sensitivity only. Forecast skill on five points cannot establish general superiority, causal mechanisms or absolute animal populations. This report and notebook concern Yash's component; the accompanying combined notebook and report integrate all five contributions.

References

[1] Bob Carpenter. Predator-prey population dynamics: the Lotka-Volterra model in Stan. Stan case study, 2018. Data file hudson-bay-lynx-hare.csv, stan-dev/example-models commit d52cde5; accessed 27 October 2026.

[2] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.